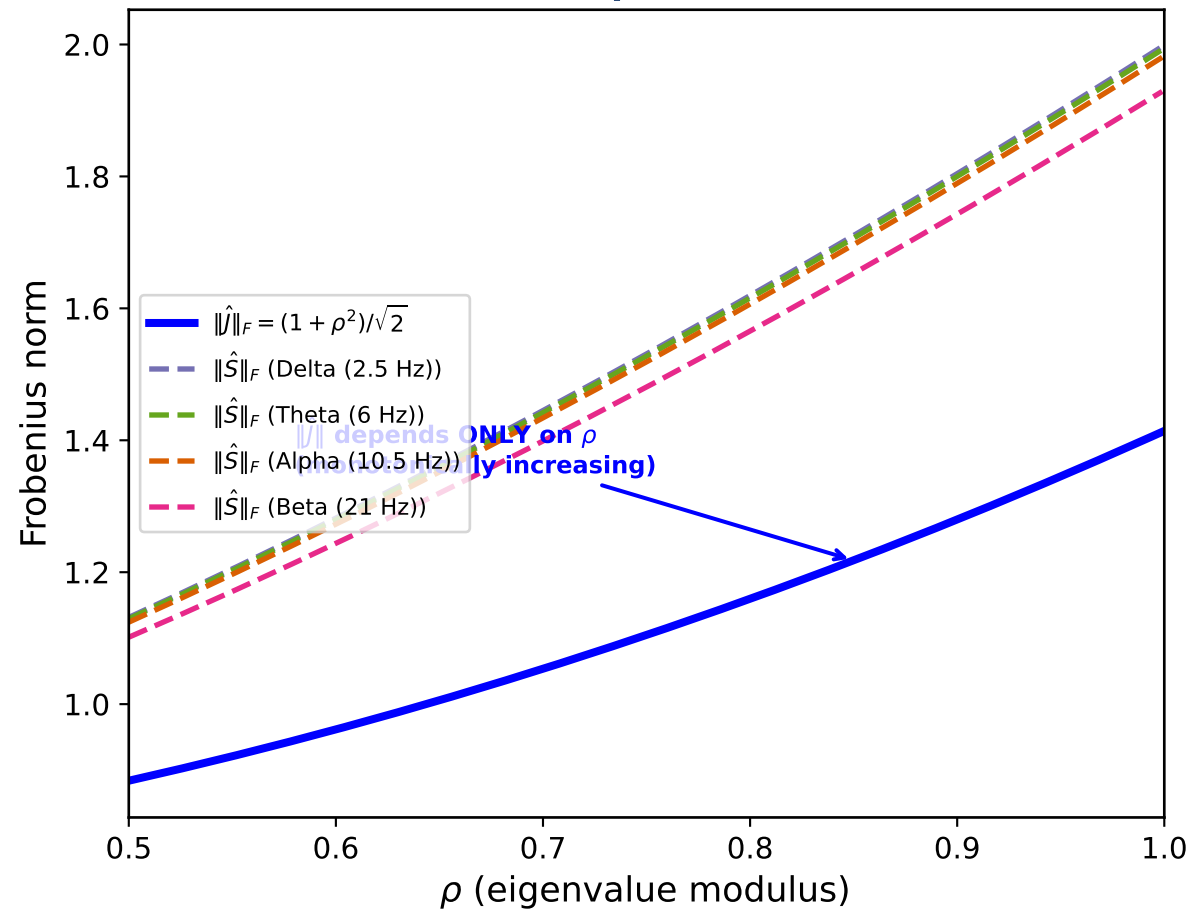
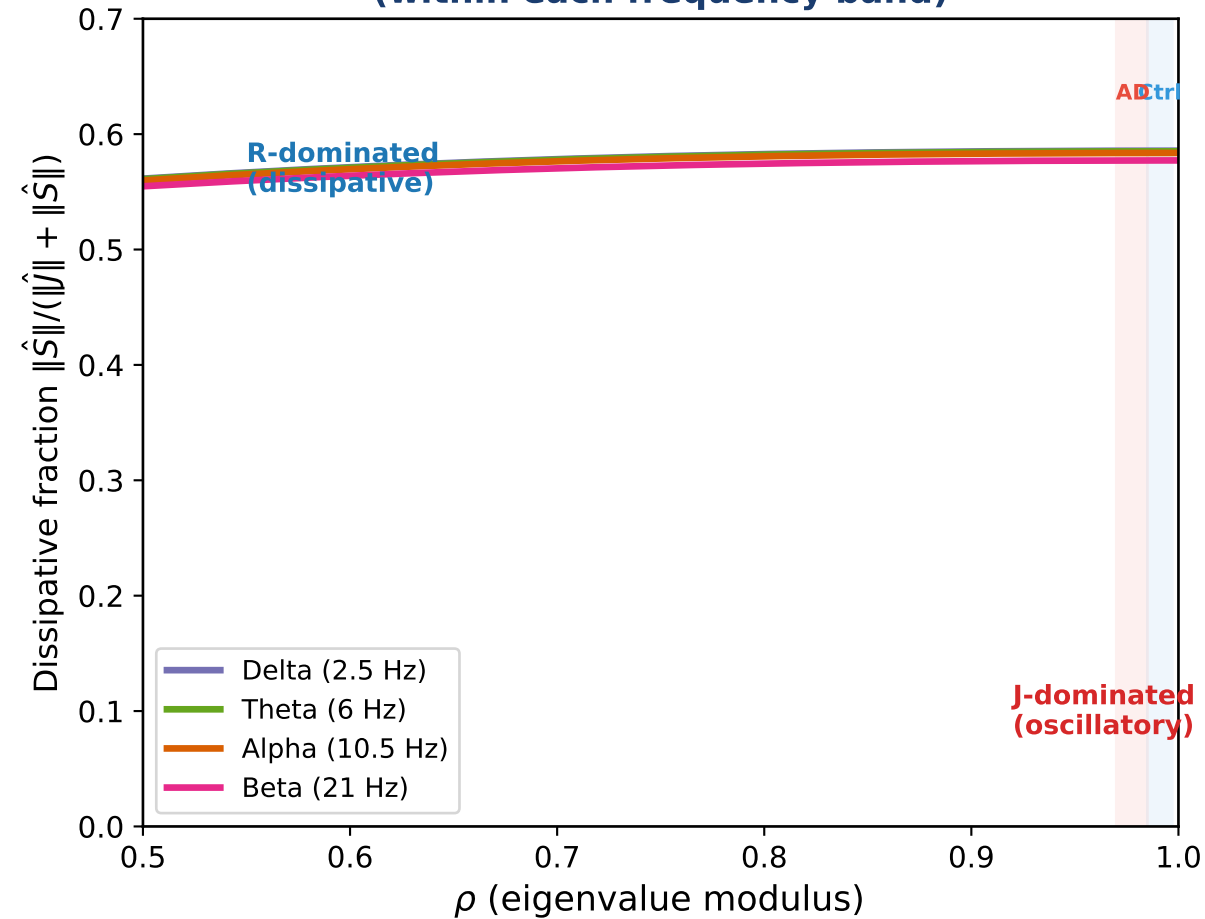


Figure 3: Port-Hamiltonian Decomposition ($\hat{J}$, $\hat{R}$) of Cortical EEG Dynamics

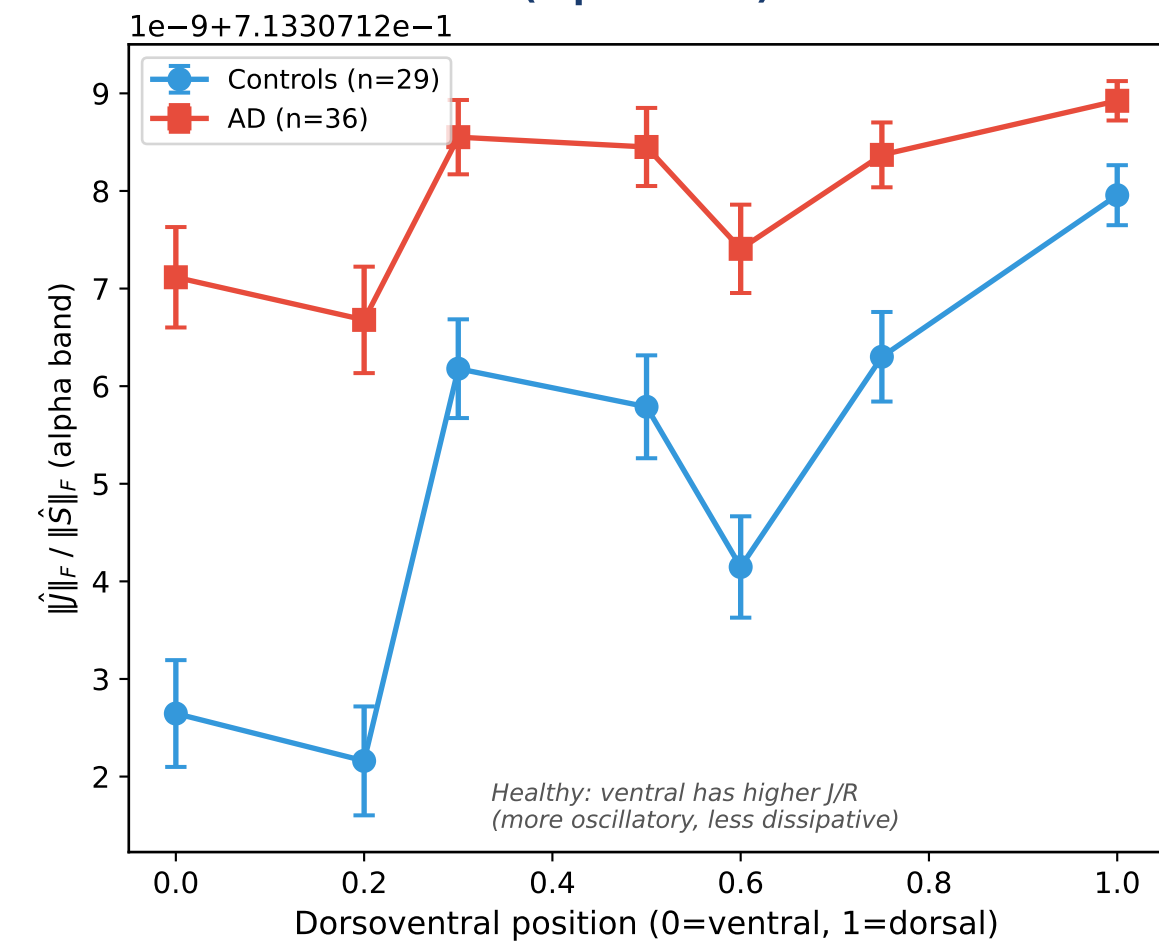
A. Analytical decomposition of the AR(2) companion matrix



B. Higher ρ = lower dissipative fraction (within each frequency band)



C. AD flattens the cortical $\hat{J}/\hat{R}$ gradient (alpha band)



D. Frequency-specific disruption of the J/R structure in AD

